



Durability performance and long-term prediction models of sand-coated basalt FRP bars

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ABSTRACT

Basalt-fiber-reinforced polymer (BFRP) bars are expected to provide benefits that are comparable or superior to other types of FRP while being significantly cost-effective. However, extensive investigations are needed to evaluate the long-term characteristics and durability performance of these bars. This article presents an experimental study that investigated the physical, mechanical, microstructural, and durability characteristics of newly developed basalt-fiber-reinforced polymer (BFRP) bars. The physical, mechanical properties and microstructural characteristics were evaluated first on the unconditioned BFRP bars. The durability performance of the BFRP bars was then assessed by conducting the mechanical tests, such as transverse-shear test, flexural test, and interlaminar-shear test, on the specimens after different exposure periods (1000; 3000; and 5000 h) at 60 °C. Thereafter, the BFRP bar properties were assessed and compared with the values obtained on the unconditioned specimens. The test parameter was conditioning time (1000; 3000; and 5000 h). The test results revealed that the unconditioned BFRP bars had the best physical properties. On the other hand, the long-term durability performance revealed that the transverse shear-strength, flexural-strength, and interlaminar shear-strength retention were decreased by 12%, 19%, and 21%, respectively.

1. Introduction

The durability and life performance of reinforced concrete structures have become a major concern for the infrastructure systems [1–3]. One of the main factors reducing durability and service life of reinforced concrete structures is the corrosion of embedded steel reinforcement bars, particularly where deicing salts are routinely used, such as in concrete deck slabs and parking garages [4–6]. Recently, fiber-reinforced-polymer (FRP) materials have emerged as an acceptable construction material, for both new constructions and rehabilitation and strengthening of existing structures, because of their noncorrodible nature, higher tensile strength, and lower weight relative to conventional steel reinforcing bars ([4,7–10,41]. FRPs are available with a wide range of mechanical properties (tensile strength, bond strength, and modulus of elasticity) and are made with high-tensile-strength fibers such as glass, carbon, and aramid embedded in polymer matrices as polyesters, vinyl esters, or epoxies. Moreover, FRPs can be produced as bars, ropes, tendons, and grids in a wide variety of shapes

and surface configurations as well as with varied characteristics ([11,41].

Recent developments in FRP technology and innovate, new types of fibers—such as basalt fibers which is made from basalt rock—are being introduced to manufacture basalt fiber-reinforced polymers (BFRPs). BFRP is the most recently FRP composite, appearing within the last decade [1,2,12–14]. BFRPs are non-corrosive, non-magnetic, possess high resistance against low and high temperatures, good chemical resistance, good environmental friendliness, high performance in terms of strength, fire resistance and durability, as well as the lower potential cost with respect to many other FRP materials. Consequently, the basalt fibers are ideally appropriate for applications involving high temperature, chemical resistance, durability, mechanical strength, and low water absorption [15]. In comparison to others FRP types, BFRP fibers lie between glass and carbon for both stiffness and strength. BFRP has higher strength and modulus, similar cost, and greater chemical stability compared with E-glass FRP. Furthermore, it offers over five times the strength and one-third the density of commonly used low-carbon

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steel bars [8,16]. In the past decade, numerous studies on the durability of glass and carbon FRP bars have been conducted which mainly comprises their performances under alkali, acid and salt environments [17–21]; and [22–25,9,26,27]; and [28]. The majority of these studies has highlighted on the short-term performance of FRP bars. Moreover, most of these studies have investigated the durability characteristics of glass FRP bars for reinforced concrete structures. Few studies, however, have assessed the physical, mechanical and durability characteristics of basalt-FRP, in order to understand their behavior and generate higher confidence in these newly developed FRP materials, under alkali and salt environments [1,2,13,14,12,16,29]. [30] investigated the applicability of the basalt, glass, and carbon fibers as a strengthening material for structural concrete members through various experimental works for durability, mechanical properties, and flexural strengthening at elevated temperatures. They reported that immersing basalt and glass fibers in an alkaline solution led to volume and strength loss through a surface reaction, whereas carbon fibers did not show any significant strength reduction. The basalt fibers, however, retained about 90% of their strength at ambient temperature after exposure at 600 °C for 2 h. Aging results indicate that the interfacial region in basalt composites might be more vulnerable to environmental damage than in glass composites. The basalt-epoxy interface, however, might also be more durable than the glass-epoxy interface in tension-tension fatigue, because basalt composites have a longer fatigue life.

An experimental study was conducted, by Ref. [31]; to determine if BFRP composites were feasible, practical, and a beneficial alternative for transportation applications. The researchers of the mentioned study concluded that the chemical composition of basalt fibers is close to glass fibers, except that the basalt contains a high ratio of iron oxide, conferring its brown color. Aging results indicate that the interfacial region in basalt composites might be more vulnerable to environmental damage than in glass composites. The basalt-epoxy interface, however, might also be more durable than the glass-epoxy interface in tension-tension fatigue, because basalt composites have a longer fatigue life. In 2008, Wang et al. investigated the chemical durability and mechanical properties of a kind of alkali-proof basalt fiber and its reinforced epoxy resin matrix composites in alkaline solution for 3 months. The basalt fiber was boiled in distilled water, sodium hydroxide and hydrochloric acid, respectively. Then the mass loss and strength change of the fibers were studied showing that the alkali resistance of the basalt fiber is better than acid resistance. The experimental results also showed that, after exposure, the modulus of the BFRP was unaffected, but its strength decreased by 40%.

In the last two years, an extensive research project has been conducted at the Department of Civil Engineering at the University of Sherbrooke to assess the short-term and long-term characteristics of newly developed BFRP bars as a preliminary step in introducing these new materials into pilot projects and FRP design codes and material specification [12,29]. [12] presented an experimental study that investigated the physical, mechanical, and durability characteristics of three different types of fiber-reinforced polymer (FRP) bars made of basalt and glass fibers with vinylester and epoxy resins. They found that the glass/vinylester composite had the best physical and mechanical properties and lowest degradation rate after conditioning in alkaline solution. The basalt/epoxy composite ranked second, while the basalt/vinylester composite evidenced the lowest physical and mechanical properties and exhibited significant degradation of its physical and mechanical properties after conditioning [29]. investigated the durability and long-term performance of BFRP bars in an alkaline solution (up to 3000 h, at 60 °C) to determine their suitability as internal reinforcement for concrete elements. It was noticed that the BFRP bars had good mechanical behavior and could be placed in the same category as grade II and grade III GFRP bars (according to tensile modulus of elasticity).

This paper represents an experimental investigation to determine the physical, mechanical, and durability characteristics of newly

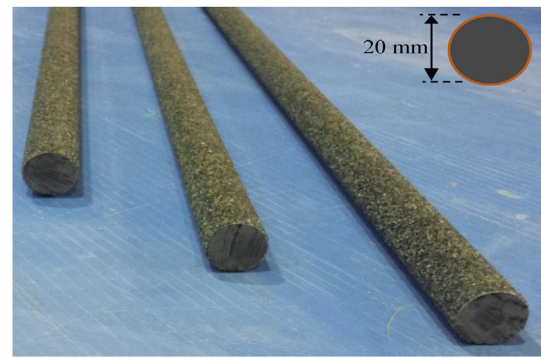


Fig. 1. BFRP bar 20 mm in diameter used in this investigation.

developed BFRP bars, manufactured by a Canadian company (Pultrall Inc.), as corrosion-resistant reinforcing material for bridge deck, girder, and pile applications in marine and aggressive environment. Since BFRPs have not been included in design standards and specifications yet, the results of this experimental study will contribute to integrating BFRP into FRP standards and guides.

2. Experimental program outline

2.1. Material properties

Sand coated basalt FRP bars manufactured by a Canadian company [32] were used in this study as shown in Fig. 1. The BFRP bars manufactured from continuous basalt fibers impregnated in vinylester resins using the pultrusion process. The nominal cross-sectional area of the bar is 284 mm², as reported by the manufacturer, while the actual (immersion) cross-sectional area is 362 mm², as specified in CSA-S806 [52]. The mechanical properties reported herein were calculated using the nominal cross-sectional area.

2.2. Test specimens

For this study, tensile [ASTM D7205 [33]], transverse shear [ASTM D7617 [33]], flexural [ASTM D4476 [34]], and short-beam shear testing [ASTM D4475 [35]] were conducted as the appropriate mechanical tests. The BFRP bars were provided into 2400; 350; and 300 mm lengths for tensile strength, transverse shear strength, and flexural properties, respectively. In addition, some specimens were cut into 120 mm lengths so that the short-beam shear test could be performed according to ASTM D4475 [35]. For tensile testing, Fig. 2b shows the dimensions of the typical tensile test specimens as specified in ASTM D7205 [45].

2.3. Conditioning of BFRP bar specimens

The basalt-FRP specimens were completely immersed in alkaline solution inside stainless steel containers specially manufactured for the study (Fig. 3). The alkaline solution was prepared using calcium hydroxide, potassium hydroxide, and sodium hydroxide (118.5 g of Ca (OH)₂ + 0.9 g of NaOH + 4.2 g of KOH in 1 L of deionized water) according to CSA S806 [52] and ACI 440.3R [38]. The pH of alkali solution was 12.8. It is worth noting that the alkaline environment in concrete has a pH above 12 (ACI 440.4R-04) [39]. The BFRP specimens and stainless steel containers were kept at 22 °C and 60 °C for 1000; 3000; and 5000 h of conditioning. The stainless steel containers were covered with polyethylene sheeting to avoid water evaporation during conditioning. Furthermore, the water level was kept constant throughout the study to avoid a pH increase that could result from decreased water level and a significant increase of alkaline ions in the solution. The immersion temperatures were chosen to accelerate the

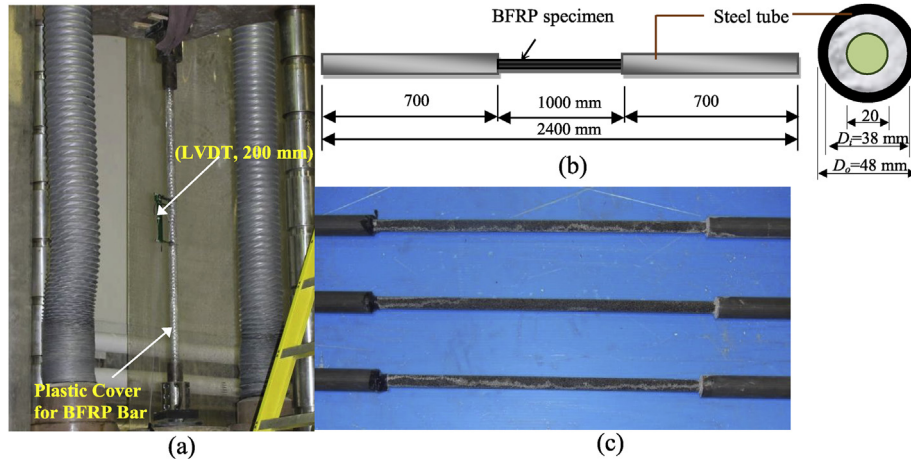


Fig. 2. Tensile test: (a) test setup; (b) specimen dimensions (mm); (b) overview of the basalt-FRP specimens attached with steel pipes.



Fig. 3. Stainless-steel container built for aging the BFRP bar specimens in alkaline solution at 60 °C.

degradation effect of aging, but they were not high enough to produce any thermal-degradation mechanisms. The BFRP specimens were removed from the alkaline solution after 1,000, 3,000, and 5000 h and tested under transverse shear, flexural, and short-beam shear to compare their strength retention values to those of the reference specimens.

3. Physical characterization of BFRP bars

The Physical and microstructural analyses were conducted on BFRP specimens. Physical properties included: (1) cross section area (2) basalt-fiber content, (3) moisture absorption, (4) cure ratio, (5) glass-transition temperature, (6) transverse coefficient of thermal expansion, and (7) wicking. These properties were determined according to CSA S807 [47]. Moreover, optical microscopy (OM), and scanning electronic microscopy (SEM) analyses were performed to investigate the material's microstructure.

3.1. Cross-sectional area

Three 280 mm long specimens were prepared and tested to determine the actual cross-sectional area of the BFRP bars as specified in CSA-S806 [52], Annex A. All BFRP specimens were kept in the test environment for 24 h prior to weighing and measuring. The actual cross-sectional area was calculated according to the following equation:

$$\text{Cross-sectional area} = 1000 \cdot (V_0 - V_1) / L \quad (1)$$

where, L is specimen length; V_0 is the volume of water added without BFRP specimen; V_1 is the volume of water added with BFRP specimen.

3.2. Fiber content

Basalt fiber content was determined by thermogravimetry according to ASTM E1131 [51]. A very small piece of material (a few tenths milligrams) was cut from the center of the bar, placed in platinum crucible and then heated up to 550 °C under inert atmosphere. The weight loss (W_L) has been recorded at a temperature equal to 550 °C. Since the material only contains basalt fibers and resin, fiber content by weight was then calculated according to the following equation:

$$\text{Fiber content by weight} = 100 \cdot (W_T - W_L) / W_T \quad (2)$$

where, W_T is the total weight before burn off.

3.3. Water-immersion test

The moisture uptake at saturation of BFRP bars was determined according to ASTM D570 [44], except that the immersions were performed in tap water instead of distilled water. Three 50-mm-long specimens were cut, dried, and weighed prior to immersion in water at 50 °C for three weeks. The samples were removed from the water after three weeks, surface dried, and weighed.

The water content at saturation in weight percent (W_s) was calculated using Equation (3).

$$W_s = 100 \cdot (P_s - P_d) / P_d \quad (3)$$

where P_s and P_d are the sample weights in saturated and dry states, respectively.

3.4. Cure ratio

Cure ratio was determined according to ASTM D5028 [43]. The enthalpy of polymerization of the sample is measured by DSC and compared to the enthalpy of polymerization of pure resin, taking into account the weight percentage of resin in the matrix. Thirty to 50 mg of sample were accurately weighed and placed in aluminum crucible. The samples were then heated from room temperature to 200 °C at a heating rate of 20 °C/min and the area of the peak of polymerization was calculated. The measurement was carried out on 3 specimens.

3.5. Glass transition temperature T_g

Glass transition temperature, T_g , was determined for 12.5 mm BFRP specimens by Differential Scanning Calorimetry (DSC) using ASTM

E1131 [51] test method. Thirty to 40 mg of composite sample were weighed and placed in an aluminum pan. The sample was then heated up to 200 °C under nitrogen at a heating rate of 20 °C/min. The value of T_g was taken at the mid-height of the heat capacity (C_p) jump.

4. Mechanical characterization of the BFRP bars

4.1. Tensile test

All BFRP bars were tested under tension according to the ASTM D7205. Each specimen was instrumented with a linear variable differential transformer (LVDT) to capture the elongation during testing. The test was carried out using a Baldwin testing machine and the load was increased until failure. Before test, steel pipes were attached to the BFRP test specimens according to CSA S806-12, Annex B “Anchor for testing FRP specimens under monotonic, sustained, and cyclic tension”. Fig. 2 shows the test setup, dimensions of test specimen, and overview of the BFRP specimens attached with steel tubes. The test specimens were instrumented with one linear variable differential transformer (LVDT, 200 mm length) to capture specimen elongation during testing. For each tensile test, the specimen was mounted in the tensile machine with the steel pipe anchors gripped by the wedges of the upper and the lower jaws of the machine. The average loading rate ranged between 250 to 560 MPa/min. The applied load and bar elongation were recorded during the test using a data acquisition system monitored by a computer. Due to the brittle nature of FRP, no yielding occurs and the stress-strain behavior was linear.

4.2. Transverse-shear test procedure

Transverse shear tests were performed according to ASTM D76147 [33] to characterize the BFRP bars. The transverse shear strengths of BFRP bar specimens were measured on samples subjected to elevated temperature (60 °C) for 1000; 3000; and 5000 h of immersion in alkaline solution. This property was investigated to provide some indications on the retention of the matrix related mechanical properties of BFRP bars subjected to high temperatures and alkaline environment. As shown in Fig. 4, the test setup consisted of a 230 × 100 × 110 mm steel base equipped with lower blades spaced at 50 mm face to face, allowing for the double transverse-shear failure of the specimen caused by an upper blade. All specimens were cut at a length of 350 mm and tested in shear according to ASTM D7617. Six unconditioned and conditioned specimens 350 mm in length were tested under laboratory conditions with an MTS 810 testing machine equipped with a 500-kN load cell. A displacement-controlled rate of 1.3 mm/min was used during the test, which yielded between 30 and 60 MPa/min until specimen failure. The loading was done without subjecting the test specimens to any shock. The transverse-shear strength of BFRP bars, τ_u , was calculated as $\tau_u = P_s/2A$, where P_s is the failure load (N); A is the cross-sectional area

(mm²).

4.3. Flexural test procedure

Flexural tests were conducted on unconditioned and conditioned BFRP specimens for 1000; 3000; and 5000 h, at 60 °C (ASTM 4476). All BFRP specimens were cut at a length of 300 mm and an overhang of 10% of the supported span was allowed at each support as shown in Fig. 5. Six unconditioned and conditioned specimens were tested under laboratory conditions using MTS 810 testing machine equipped with 500 kN load cell. The specimens were loaded at the midspan with a circular nose; the specimen ends rested on two circular supports that allowed the specimens to bend. A displacement-controlled rate of 3.0 mm/min was used during the test. The rate of loading was done without subjecting the test specimen to any shock. The applied load and deflection were recorded during the test using a data acquisition system monitored by a computer. The flexural strength of the BFRP bars, f_u , was determined as $f_u = PLC/(4I)$, where P is the failure load, L is the clear span, C is the distance to the centroid of the extreme-most fibers, and I is the moment of inertia.

4.4. Short-beam shear test procedure (interlaminar shear test)

In pultruded FRP bars, fibers are arranged unidirectionally and bonded with the polymer matrix, the horizontal stresses would be more conducive to inducing interface degradation than transverse-shear stresses [42,46]. The shear test was conducted on six specimens of BFRP bar according to ASTM D4475 [35] in order to calculate the interlaminar shear strength, which is governed by the fiber/matrix interface. The tests were carried out with a 500-kN MTS 810 testing machine. The distance between the shear planes was set to 6 times the nominal diameter of the FRP bars. Fig. 6 shows the test setup and typical modes of failure of the tested specimens. A displacement-controlled rate of 1.3 mm/min was employed during the test. The applied load was recorded with a computer-monitored data acquisition system. The interlaminar shear strength, S_u , of the BFRP bars was determined as $S_u = 0.849P/t^2$, where S_u interlaminar-shear strength (MPa); P = shear failure load (N); and t = thickness of BFRP specimen (mm).

5. Tests results and discussion

5.1. Physical properties and SEM results

Table 1 presents the physical properties of the unconditioned basalt-FRP bars, where the glass transition temperature (T_g) was determined with differential scanning calorimetry (DSC) [ASTM D3418 [36]] (see Fig. 7). From Table 1, the test results indicated that the basalt-fiber content in weight of basalt-FRP bars was 81.0%. The mass percentages of water uptake after 24 h, 7 days and at saturation were found to be

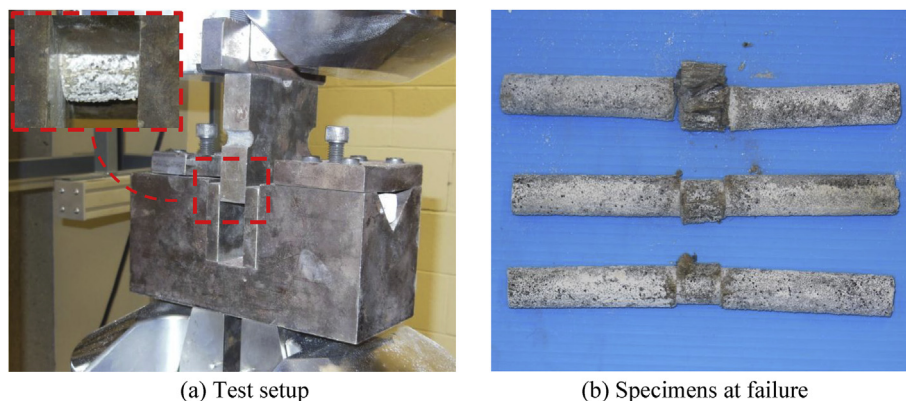


Fig. 4. Transverse shear test setup and specimens at failure.

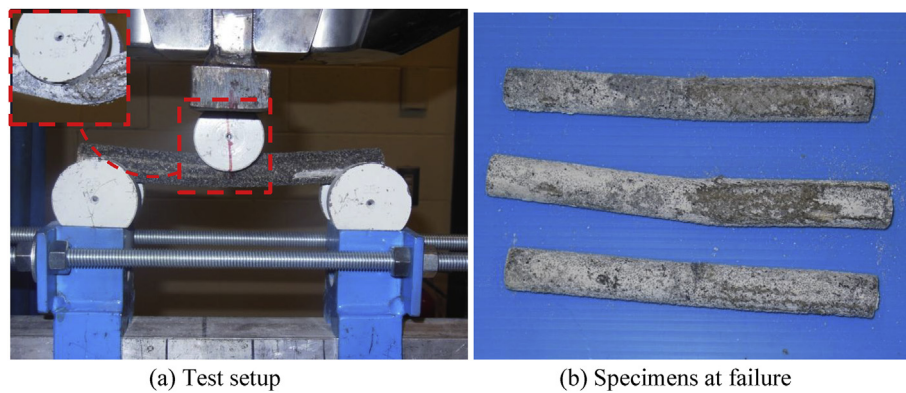


Fig. 5. Typical flexural test setup and specimens at failure.

0.04%, 0.18, and 0.25% on average. The water-absorption values obtained not exceed the limits specified in CSA S807 [47] and ACI 440 [40] (1%). The material's cure ratio is high (100%). Table 1 presents, also, the T_g values for the first and second heating of unconditioned samples (reference samples). Note that, for the unconditioned specimen, the T_g corresponding to the second heating run was close to the T_g corresponding to the first scan. These results confirm the high cure ratio of more than 99% measured by DSC and shown in Table 1.

Scanning electron microscopy (SEM) observations and image analysis were performed to observe the microstructure of BFRP specimens. All specimens observed in the SEM were first cut, polished, and coated with a thin layer of gold-palladium by a vapor-deposit process. After coating the surfaces, microstructural observations were performed on a JEOL JSM-840 A SEM. These observations were conducted to see any defect in polymer matrix, basalt fibers, or interfaces, if any. A general view of basalt-FRP bar was obtained as shown in Fig. 8. No pore or void was found in a SEM micrograph at the interfaces between the fibers and matrix. It has to be noted that the material contains fibers with various diameters as shown in Fig. 9. SEM micrograph of interface between basalt and resin matrix in Fig. 10 clearly shows that no defect in polymer matrix, basalt fibers, or interfaces was observed. The bonding between the basalt fibers and the thermosetting resin is good since there were no free gaps at the interface.

6. Mechanical properties results

6.1. Tensile strength, modulus of elasticity, and strain

The tensile test of unconditioned BFRP specimens showed an approximately linear behavior up to failure. Specimens failed through rupture of fibers. The failure was accompanied by delamination of fibers and resin, as shown in Fig. 11. Table 2 presents the experimental results obtained during the tensile tests concerning the ultimate tensile

Table 1

Physical properties of the reference basalt FRP bars.

Property	Basalt FRP bars	Specified limit	
		ACI 440.1R-15	CSA S807-10
Fiber content by weight (%)	81	55% (by vol.)	70% (by wt.)
Cure ratio (%)	100	N/A	93 (D2); 95 (D1)
Transverse CTE, ($\times 10^{-6} \text{C}^{-1}$)	22	N/A	40
Glass transition temperature (T_g)	Run 1 116 Run 2 117	100	80 (D2); 100 (D1)
Moisture uptake (%)	0.25	< 1.0	1.0 (D2); 0.75 (D1)
Diameter (mm)	20.0	N/A	N/A
Cross-sectional area (mm^2)	362	N/A	N/A
Wicking dots	0.0	N/A	N/A

*Note: CTE = coefficient of thermal expansion. D1 and D2 classifications can be found in CSA S807-10.

*The mechanical properties were calculated using the nominal cross-sectional area.

strength, modulus of elasticity, and ultimate strain of unconditioned BFRP bars. As shown in Table 2, the tensile strength, modulus of elasticity, and ultimate strain of unconditioned BFRP bars were equal to 1646 ± 12 MPa; 69.7 GPa; and 2.4%, while these values, as reported by the manufacturer, are 1636 MPa; 64 GPa; and 2.5%, respectively. Fig. 11 shows the mode of failure during the tensile tests for unconditioned (reference) BFRP bars.

6.2. Effect of conditioning on transverse-shear strength

Table 2 shows that the transverse-shear strength of the

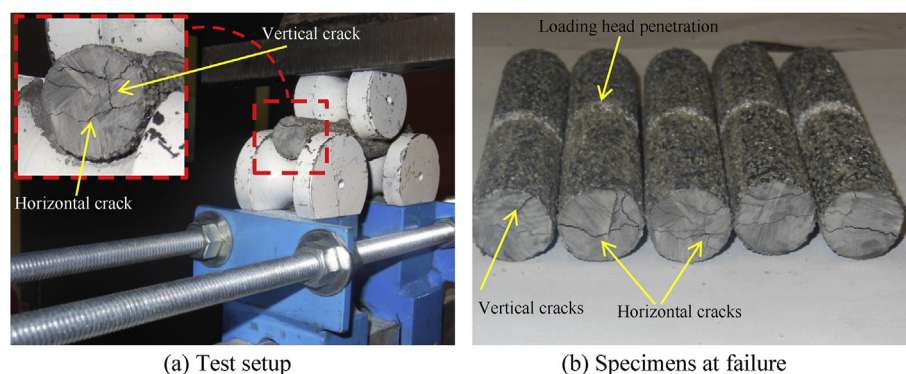


Fig. 6. Interlaminar-shear test setup (short-beam test) and specimens at failure.

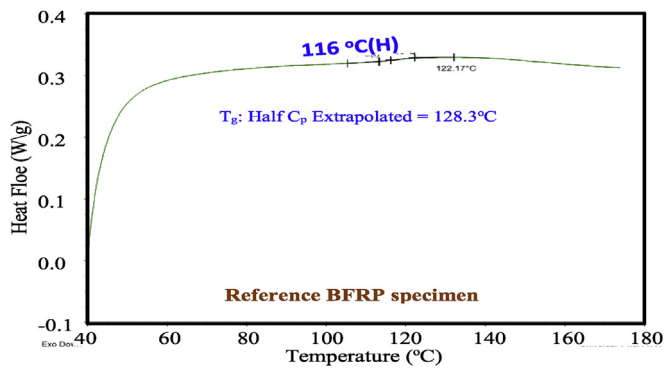


Fig. 7. DSC scans for glass transition temperature (T_g).

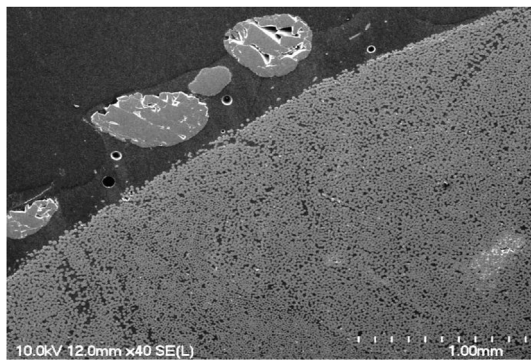


Fig. 8. Micrographs of the BFRP bars cross-section (20 mm-diameter).

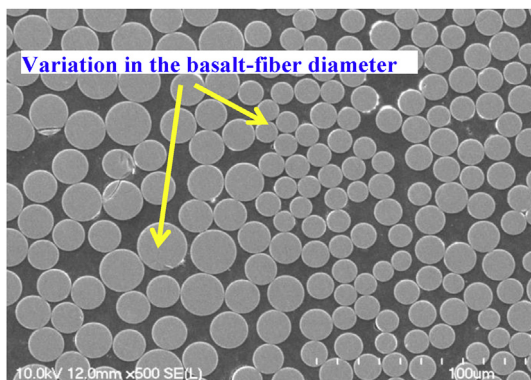


Fig. 9. Micrographs of the basalt FRP bar cross section at medium magnification.

unconditioned basalt-FRP bars was 241 ± 2.1 MPa, while Table 3 shows the transverse-shear strength and strength-retention ratio of the tested BFRP bars after 1,000, 3,000, and 5000 h of immersion in the alkaline solution at 60 °C. Table 3 indicates that the BFRP bars were affected by accelerated aging with a transverse-shear strength reduction of 12% after 5000 h of immersion. Fig. 12 shows the effect of the alkaline solution on the transverse shear strength after different exposure times. The BFRP bars exhibited no significant reductions in the early stages (less than 3000 h).

6.3. Effect of conditioning on flexural strength

The three-point flexural strength, flexural modulus of elasticity, and ultimate outer-fiber strain of the tested unconditioned BFRP bars, are presented in Table 2. The flexural strength and flexural modulus of elasticity were 653 ± 15 MPa and 41.0 GPa, respectively. Table 3 and Fig. 13a show the experimental results obtained during the flexural

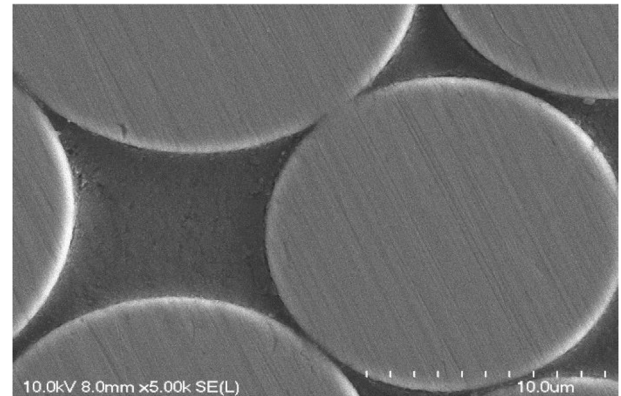
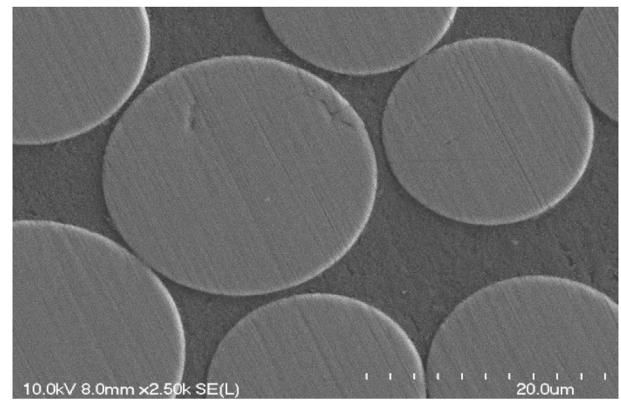


Fig. 10. Micrographs of the bars for basalt fiber/matrix interface.



Fig. 11. Typical failure mode of tested BFRP specimens subjected to tensile test.

Table 2
Mechanical properties of the reference basalt FRP bars.

Property	BFRP bars	Specified limit	
		ACI 440.1R-15	CSA S807-10
Transverse shear strength (MPa)	241 ± 2.1	> 124 MPa	> 160 MPa
Tensile strength (MPa)	1646	–	–
Tensile modulus of elasticity (GPa)	69.7	39.3 GPa	40.0 GPa
Ultimate strain (%)	2.4	> 1.2%	> 1.2%
Short-beam shear strength (MPa)	30 ± 0.5	–	–
Four-point flexural strength (MPa)	653 ± 15	N/A	N/A
Flexural modulus (stiffness) (GPa)	77.2 ± 0.3	N/A	N/A

*The mechanical properties were calculated using the nominal cross-sectional area.

Table 3
Retention of mechanical properties of the conditioned BFRP bars.

Fiber/resin	Conditioned period	Temp. (°C)	τ_u (MPa)	Ret. (%)	f_u (MPa)	Ret. (%)	E (GPa)	Ret. (%)	S_u (MPa)	Ret. (%)
Basalt-FRP bars	1000	22	240	99.6	647	99.1	77.0	99.7	29.5	98.3
		40	237	98.3	646	89.9	76.8	99.5	28.2	94.0
		60	233	96.7	642	98.3	75.8	98.2	28.0	93.3
	3000	22	239	99.1	594	90.9	76.6	99.2	27.9	93.0
		40	233	96.7	592	90.6	76.7	99.3	27.3	91.0
		60	230	95.4	590	90.3	73.9	95.7	27.0	90.0
	5000	22	227	94.2	540	83.0	76.2	98.7	25.8	86.0
		40	222	92.1	533	81.6	76.1	98.5	25.3	84.3
		60	214	88.8	528	80.0	70.8	91.7	24.1	80.0

Note: τ_u = Transverse shear strength (MPa); f_u = Four-point flexural strength (MPa); S_u = Short-beam shear strength (MPa); Ret. = Retention of strength (%).

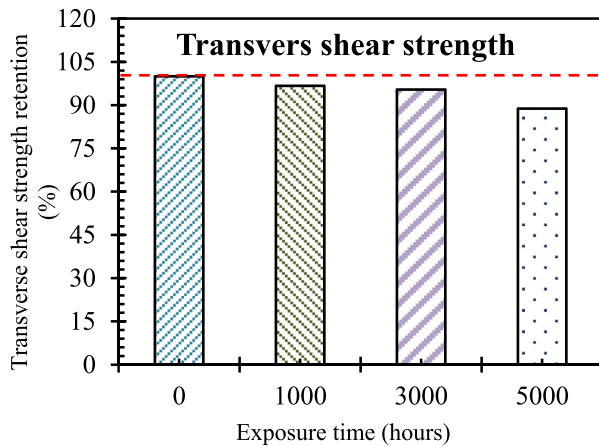


Fig. 12. Effect of conditioning in the alkaline solution at 60 °C on transverse-shear strength.

tests of aged BFRP bars tested after immersion in alkaline solution. Fig. 13a shows flexural strength loss observed with BFRP reinforcing bars conditioned in alkaline solution during 5000 h (208 days). The BFRP bars had flexural-strength reductions of 2% and 10% after 1000 h, and 3000 h, while the bars showed a lower reduction of 19% after 5000 h. These observations confirm that the bond between the BFRP fibers and resin after conditioning was lower than that before conditioning. Consequently, debonding occurring at the fiber–matrix interface caused the fibers to separate from the resin. Fig. 13a shows the effect of the alkaline solution on flexural strength. Fig. 5b shows the mode of failure of failure during the flexural tests for reference and specimen aged at 60 °C during 5000 h. It can be seen in Fig. 5a that the

tested specimen was loaded until rupture occurs in the extreme tensile fibers and that aged and reference BFRP bars have similar mode of failure. Note that the extreme compressive fibers did not show any premature compression failure. Progressive damage propagation occurred through matrix cracking, fiber breaking, and interfacial debonding of the extreme tensile fibers side of specimens.

Table 3 gives, also, the flexural modulus of elasticity and the retention ratio of the tested BFRP bars after 1,000, 3,000, and 5000 h of immersion. The bars had no significant differences in flexural modulus of elasticity after 5000 h. The reduction ranged from 3% to 11% in comparison to the references. Fig. 13b illustrates the effect of the alkaline solution at 60 °C on the flexural modulus of elasticity, with all bar specimens exhibiting a steady reduction rate.

6.4. Effect of conditioning on interlaminar-shear strength

The short-beam shear test revealed that the unconditioned BFRP bars had the interlaminar-shear strength of 30.2 ± 2.7 MPa, (Table 2). It is worth mentioning that the high values of the interlaminar-shear strength reveal a strong interface between the resins and reinforcing fibers, which was clarified in the SEM analysis of the unconditioned BFRP specimen.

Table 3 also shows the apparent horizontal shear (interlaminar shear) strength and strength-retention ratios of the tested FRP bars after 1,000, 3,000, and 5000 h of immersion in alkaline solution. The reduction ratio for the BFRP bars after 5000 h was 21%. Fig. 14 shows the effect of the alkaline solution on the interlaminar-shear strength, with the BFRP bars, for different time of immersions.

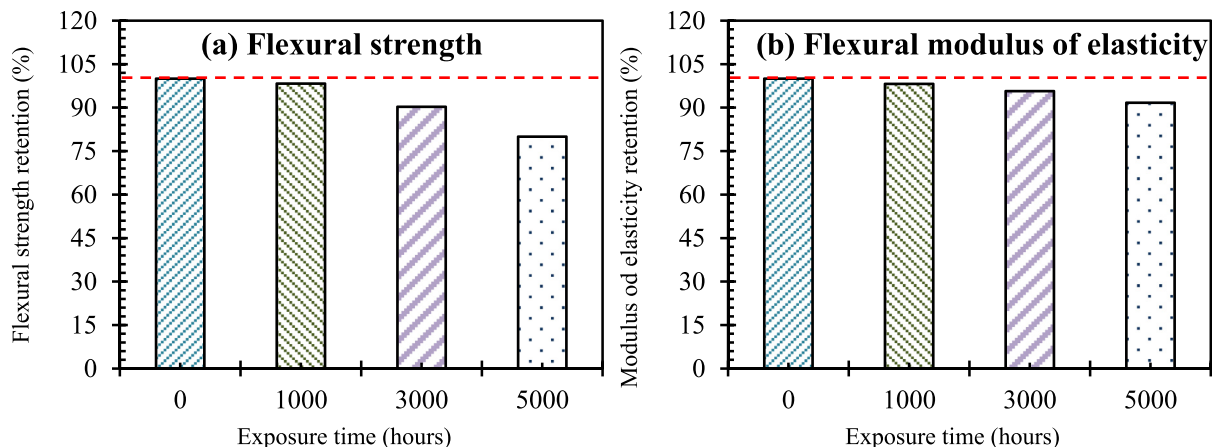


Fig. 13. Effect of conditioning in the alkaline solution at 60 °C on mechanical properties: (a) flexural strength; (b) flexural modulus of elasticity.

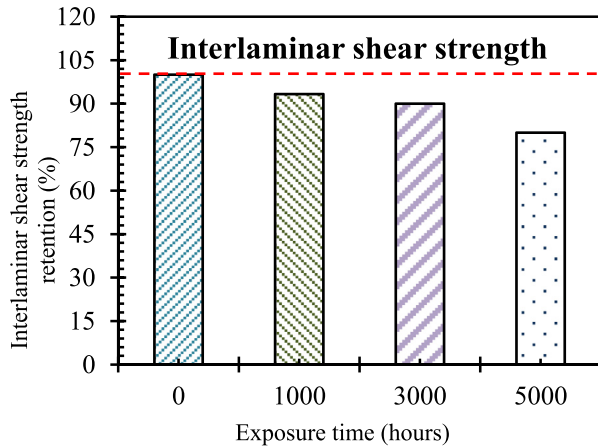


Fig. 14. Effect of conditioning in the alkaline solution at 60 °C on interlaminar-shear strength.

7. Prediction of long-term behavior and service life for transverse shear strength

7.1. Arrhenius model

Several researchers have been employed the Arrhenius concept for predicting the long-term performance of FRP (Bank et al., 2003; Robert et al., 2013, 2009; [9,21,22,26]. Eq. (4) expresses the Arrhenius relation-ship, in terms of the degradation rate (Nelson, 1990).

$$k = Ae^{\left(\frac{-E_a}{RT}\right)} \quad (4)$$

where k = degradation rate; A = constant relative to the material and degradation process; E_a = reaction's energy of activation; R = universal gas constant; and T = temperature in °C. The primary assumption with this model is that only one dominant degradation mechanism of the material operates during the reaction and that this mechanism will not change with time and temperature during the exposure [22]. Only the rate of degradation will be accelerated as the temperature increases.

Dejke (2001) used another approach to generate a relative TSF. He proposed using the TSF to transform the time in the accelerated test to actual service lives for GFRP reinforcement. Because the time for a certain reaction to take place must be proportional to the inverse of the rate of reaction, Dejke (2001) proposed determining the TSF in accordance with Eq. (5).

$$TSF = \frac{t_1}{t_2} = e^{B\left(\frac{1}{T_1+273.15} - \frac{1}{T_2+273.15}\right)} \quad (5)$$

where T_1 and T_2 = exposure temperatures (°C); and t_1 and t_2 = times

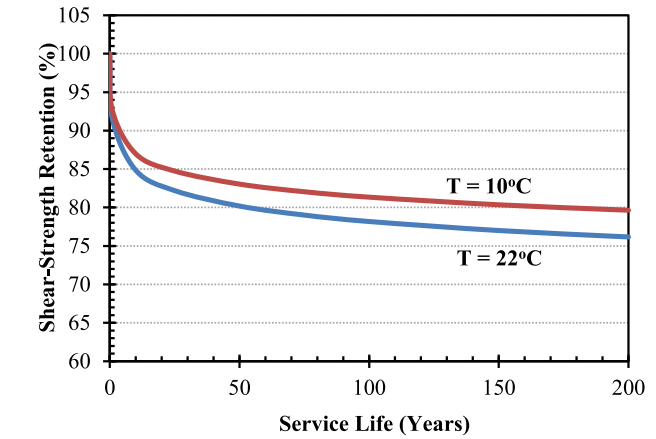
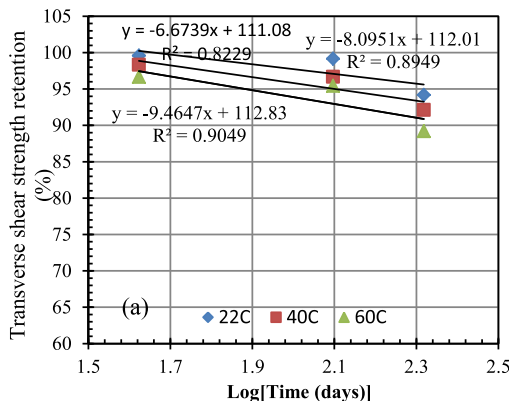


Fig. 16. General relation between the PR and the predicted service life at mean annual temperatures of 10 and 22 °C.

required to obtain a certain level of decrease in mechanical property at temperatures T_1 and T_2 , respectively. The TSF is sensitive to activation energy, and a good estimate of E_a is needed to generate a reasonable TSF. According to the procedure proposed by Ref. [48,49], the natural logarithm of time to reach a set of levels of normalized performances versus $1/T$, expressed as the inverse of absolute temperature (1000/K), was used to predict the service life of the BFRP bars at mean annual temperatures (MAT) of 10 °C. The temperature of 10 °C corresponds well to the mean average temperature of northern regions, where deicing salts are often used. A coefficient of determination (R^2) value close to 1 was desired. ASTM procedures, however, recommend a minimum value of 0.80 for acceptability; the obtained R^2 values were between 0.96 and 0.99. The service-life time necessary to reach the established tensile-strength retention levels (PR) can be extrapolated for any temperature from the Arrhenius plot. Consequently, predictions were made for transverse shear-strength retention as a function of time for immersion at 10 °C, and the general relation between the PR and the predicted service life at the average temperature of 10 °C can be drawn (Fig. 16). The predicted time to reach the determined tensile-strength retention level (PR) for the BFRP bars aged in an alkaline solution simulating the concrete environments at an isotherm temperature of 10 °C is approximately 150 years for a PR of more than 80%. Moreover, the predicted service life of the BFRP bars aged in the alkaline solution at an isotherm temperature of 10 °C with a PR of less than 80% can be estimated as being infinity. These predictions show that the BFRP bars tested in this study are a high durability in a concrete environment. On the other hand, the prediction curves for the BFRP bars predicted transverse shear-strength retention of 82% after a service life of 75 years. Table 4 presents the transverse shear-strength retention after 10, 25, 50, 75, and 100 years of service at MATs of 10 °C and 30 °C. Table 4

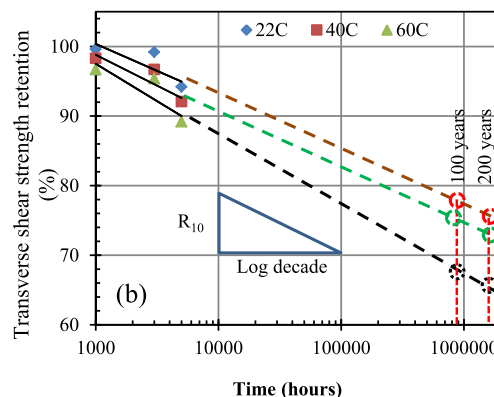


Fig. 15. Plot of the transverse shear strength retention of BFRP bars as a function of time.

Table 4

Transverse shear-strength retention at different service-life periods at MATs of 10 and 30 °C based on Arrhenius model.

Service life (Years)	Transverse shear-strength retention (%)	
	10 °C	30 °C
10	86.9	84.9
25	84.7	82.1
50	83.0	80.2
75	82.0	79.0
100	81.3	78.1
150	80.2	77.0
200	79.6	76.1

indicates that, even after a service life of 100 years, which corresponds to the maximum expected service life, the shear-strength retention was still 81.3%, for BFRP bars. At 30 °C, the shear-strength retention of BFRP bars was 76.1, 77.0, 78.1, and 79.0% after a service life of 200, 150, 100, 75 years, respectively, (Table 4).

7.2. Fib bulletin (40) method

In 2007, FIB Bulletin (40) Task Group 9.3 proposed durability design approach for FRP bars by incorporating factors of relative humidity (RH), exposed mean average temperature (MAT), and services life (FIB, 2007). The long-term design transverse shear-strength is given by Eq. (6)

$$f_{fd} = \frac{f_{fko}}{(\eta_{env,t} \cdot \gamma_f)} \quad (6)$$

where f_{fd} is a long-term design value of transverse shear-strength for BFRP bars; f_{fko} is the characteristic value of transverse shear-strength (short term), which the composite bars can resist after exposure to a practical test environment for 1000 h and this value can be expressed as a percentage of the transverse shear-strength or as absolute value; γ_f is material factor; $\eta_{env,t}$ is environmental strength reduction factor which is obtained by Eq. (7)

$$\eta_{env,t} = \frac{1}{[(100 - R_{10})/100]^n} \quad (7)$$

where R_{10} is the standard reduction in transverse shear-strength in percent per decade (logarithmic decade) due to environmental effects, which can be extrapolated from each individual degradation line (Fig. 15). The exponent n is the sum of four influence terms as shown in Eq. (8)

$$n = n_{mo} + n_T + n_{SL} + n_d \quad (8)$$

Table 5

Transverse shear-strength retention predications after service life of 100 years based on the method in fib Bulletin 40.

Material	Moisture condition	n_{mo}	MAT (oC)	n_T	n_{SL}	n	$\eta_{env,b}$	$1/\eta_{env,b}$
Basalt-FRP bars (12 mm in diameter)	Dry (RH = 50%)	-1	< 5	-0.5	3.0	1.5	1.136	87.9%
			5–15	0.0	3.0	2.0	1.185	84.3%
			15–25	0.5	3.0	2.5	1.237	80.8%
			25–35	1.0	3.0	3.0	1.291	77.4%
	Moist (RH = 80%)	0	< 5	-0.5	3.0	2.5	1.237	80.8%
			5–15	0.0	3.0	3.0	1.291	77.4%
			15–25	0.5	3.0	3.5	1.347	74.2%
			25–35	1.0	3.0	4.0	1.406	71.1%
	Moisture saturated (RH = 100%)	+1	< 5	-0.5	3.0	3.5	1.347	74.2%
			5–15	0.0	3.0	4.0	1.406	71.1%
			15–25	0.5	3.0	4.5	1.407	68.1%
			25–35	1.0	3.0	5.0	1.531	65.3%

*Note that: $R_{10} = 8.17\%$ and $n_{SL} = 3.0$.

where n_{mo} , n_T , n_{SL} , n_d are the influence terms for moisture condition, temperature, desired service life, and diameter influence, respectively.

It can be seen that an exponential approach is used in the design strength equation, for which the reason is because the deterioration is described best by the kinetics of the chemical and physical processes (Weber, 2006). This assumption itself is reasonable; however, as can be seen, the base of the exponential in expressions (6) and (7) is the strength retention $(1 - R_{10})$, other than the strength reduction R_{10} . As known, the kinetics of the chemical and physical processes is linked to the outer layer of BFRP bar that degrades by the environmental attacks. Hence, the power terms n , including the effects of moisture, temperature, time, and diameter, should be a link to the outer degraded part of BFRP bar reflected as R_{10} . Thus, the design strength equation in the FIB (2007) approach is of low confidence also.

The values of the environmental influence parameter R_{10} of the BFRP specimens conditioned at 22 °C, 40 °C, and 60 °C were 7.3%, 7.7%, and 9.5%. From the curve fitting, the shear-strength retentions after 100 years were 78%, 76%, and 68% at 22 °C, 40 °C, and 60 °C, respectively. According to fib [37]; for instance, $n_{mo} = 1$ and $n_{SL} = 3.0$ at a service life of 100 years, assuming a moisture-saturated condition. As adopted by Ref. [50], n_T is equal to 0.5, 1.5, and 2.5 at 22 °C, 40 °C, and 60 °C, respectively. The value R_{10} for all the environments tested can be determined by using the average slope of the individual degradation lines, assuming that the degradation rate is similar regardless of environment [50]. Thus, R_{10} is equal to 8.17%. The estimated shear-strength retention $(1/\eta_{env,b})$ is equal 68.1%, 62.6%, and 57.5% at 22 °C, 40 °C, and 60 °C, respectively. Noticeable differences between the two methods were observed for each environmental conditioning. The differences might be attributed to increased concrete strength resulting from immersion. This is not considered in the equation, nor are the effects of moisture diffusion on the degradation mechanism. Table 5 reports the $(1/\eta_{env,b})$ predications at different moisture-saturated conditions and mean annual temperatures (MATs) after 100 years of service life. The shear-strength retentions after 100 years of service life in dry, moist, and moisture saturated environments and MATs varied from 65% to 88%, as reported on Table 5. In order to validate the FIB Bulletin (40) model's reliability, further work is needed with different accelerated moist environments, different service life influence, and longer exposure times.

8. Summary and conclusions

In this research study, new basalt-FRP bars were exposed to an alkaline solution simulating a concrete environment to determine the bars' suitability as internal reinforcement for concrete elements. Physical, mechanical, microstructural analyses and durability characterization were conducted. The BFRP specimens were immersed in

alkaline solution (1000; 3000; and 5000 h) and subjected to elevated exposure temperature (60 °C) to simulate the alkaline effect of concrete. In addition, differential-scanning calorimetry (DSC) and scanning electron microscopy (SEM) were used to characterize the physical properties of BFRP bar specimens. Based on the results of this study, the following conclusions may be drawn on the tested products:

- 1 The test observation indicated that the basalt fiber content is 81% by weight and the water uptake at saturation is equal to 0.25%. The cure ratio of the material is very high (close to 100%) but its glass transition temperature is 116 °C (H) by DSC.
- 2 Optical microscopy and electronic scanning microscopy (SEM) analysis of the unconditioned BFRP bars showed that no defect in polymer matrix, basalt fibers, or interfaces was observed. The bonding between the basalt fibers and the thermosetting resin was good since there was no free gap at the interface.
- 3 The results indicate that the transverse-shear strength of the BFRP specimens was slightly affected by increasing the immersion duration at higher temperature levels. After 5000 h of immersion in the alkaline solution at 60 °C, test result indicated that 12% degradation in the transverse-shear occurred.
- 4 The flexural strength of the BFRP bars was significantly affected by accelerated aging (19% reduction after 5000 h, at 60 °C).
- 5 The interlaminar-shear strength of the BFRP bars was highly affected by accelerated aging (21% reduction after 5000 h, at 60 °C). The fiber–resin interface plays a significant role in controlling the degradation due to conditioning.
- 6 According to the long-term predictions, the transverse shear-strength retention of the BFRP bars immersed in the alkaline solution will decrease by 19.8% and 23.0% after 150 years at an isotherm temperature of 10 °C and 30 °C, respectively. It was shown that the BFRP bars service life with a transverse shear-strength retention of less than 79.6 and 76.1% at 10 °C and 30 °C, respectively, should be infinite.

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References

- [1] Wang J, GangaRao H, Liang R, Liu W. Durability and prediction models of fiber-reinforced polymer composites under various environmental conditions: a critical review. *J Reinforc Plast Compos* 2016;35(3):179–211.
- [2] Wang X, Zhao X, Wu Z, Zhu Z, Wang Z. Interlaminar shear behavior of basalt FRP and hybrid FRP laminates. *J Compos Mater* 2016;50(8):1073–84.
- [3] Ali AH, Mohamed, Benmokrane B, El-Safty A. Effect of applied sustained load and severe environments on durability performance of carbon-fiber composite cables (CFCCs). *J Compos Mater* 2018;1–16 <https://doi.org/10.1177/0021998318789742>.
- [4] Wang J, GangaRao H, Liang R, Zhou D, Liu W, Fang Y. Durability of glass fiber-reinforced polymer composites under the combined effects of moisture and sustained loads. *J Reinforc Plast Compos* 2015;34(21):1739–54.
- [5] Ali AH, Benmokrane B, Mohamed HM, Manalo A, El-Safty A. Statistical analysis and theoretical predictions of the tensile strength retention of GFRP bars based on resin degradation. *J Compos Mater* 2018;52(21):1–20 <https://doi.org/10.1177/0021998318755866>.
- [6] Ali AH, Mohamed HM, Chaallal O, Benmokrane B, Ghrib F. Shear resistance of RC circular members with FRP discrete hoops versus spirals. *Eng. Struct. J.* 2018;174:688–700.
- [7] Ali AH, Mohamed HM, Benmokrane B. Shear strength of circular concrete beams reinforced with glass fiber-reinforced polymer bars and spirals. *ACI Struct J* 2017;114(1):39–49.
- [8] Arias JPM, Vazquez A, Escobar ME. Use of sand coating to improve bonding between GFRP bars and concrete. *J Compos Mater* 2012;46(18):2271–8.
- [9] Benmokrane B, Ali AH, Mohamed H, Robert M, ElSafty A. Durability performance and service life of CFCC tendons exposed to elevated temperature and alkaline environment. *J Compos Construct* 2016;04015043 [https://doi.org/10.1061/\(ASCE\)CC.1943-5614.0000606](https://doi.org/10.1061/(ASCE)CC.1943-5614.0000606).
- [10] Benmokrane B, Mohamed HM, Ali AH. Service-life-prediction and field application of GFRP tubular and solid bolts based on laboratory physical and mechanical assessment. *J Compos Mater* 2018;1–15 <https://doi.org/10.1177/0021998318764806>.
- [11] Ali AH, Mohamed HM, Benmokrane B, ElSafty A. Influence of resin type on physical, mechanical and durability performance of glass-FRP bars in concrete environment at elevated temperature. Fifth international conference on durability of fibre reinforced polymer (FRP) composites for construction and rehabilitation of structures (CDCC 2017), 19–21th July, 2017, Sherbrooke, QC, Canada. 2017.
- [12] Benmokrane B, Elgabbas F, Ahmed E, Cousin P. Characterization and comparative durability study of glass/vinylester, basalt/vinylester, and basalt/epoxy FRP bars. *J Compos Construct* 2015;04015008 [https://doi.org/10.1061/\(ASCE\)CC.1943-5614.0000564](https://doi.org/10.1061/(ASCE)CC.1943-5614.0000564).
- [13] Shen D, Ojha B, Shi X, Zhang H, Shen J. “Bond stress–slip relationship between basalt fiber-reinforced polymer bars and concrete using a pull-out test. *J Reinforc Plast Compos* 2016;35(9):747–63.
- [14] Wang X, Wu G, Wu Z. Evaluation of prestressed basalt fiber and hybrid fiber reinforced polymer tendons under marine environment. *Mater Des* 2014;64:721–8. 9.
- [15] INFOMINE Research Group. Basalt fiber-based thermal insulating materials market research in Russia, Moscow. 2007.
- [16] Zhishen W, Xin W, Gang W. Advancement of structural safety and sustainability with basalt fiber reinforced polymers. *Proc. Of 6th international Conference on FRP Composites in Civil engineering (CICE)*, Rome, Italy, IIFC, 2012. 2012. p. 29.
- [17] Davalos JF, Chen Y, Ray I. Long-term durability prediction models for GFRP bars in concrete environment. *J Compos Mater* 2011;46(16):1899–914.
- [18] Ng SC, Lee S. A study of flexural behavior of reinforced concrete beam strengthened with carbon fiber-reinforced plastic (CFRP). *J Reinforc Plast Compos* 2002;21:919–38.
- [19] Taljsten B, Carolin A, Nordin H. Concrete structures strengthened with near surface mounted reinforcement of CFRP. *Adv Struct Eng* 2003;6:201–13.
- [20] Won JP, Park CG. Effect of environmental exposure on the mechanical and bonding properties of hybrid FRP reinforcing bars for concrete structures. *J Compos Mater* 2006;40(12):1063–76.
- [21] Chen Y, Davalos JF, Ray I. Durability prediction for GFRP bars using short-term data of accelerated aging tests. *J Compos Construct* 2006;10(4):279–86.
- [22] Chen Y, Davalos JF, Ray I, Kim HY. Accelerated aging tests for evaluation of durability performance of FRP reinforcing bars for concrete structures. *Compos Struct* 2007;78(1):101–11.
- [23] Davalos JF, Chen Y, Ray I. Long-term durability prediction models for GFRP bars in concrete environment. *J Compos Mater* 2012;46:1899–914.
- [24] Karbhari VM, Stachowsky C, Wu L. Durability of pultruded E glass/vinylester under combined hygrothermal exposure and sustained bending. *J Compos Construct* 2007;19(8):665–73.
- [25] Robert M, Wang P, Cousin P, Benmokrane B. Temperature as an accelerating factor for long term durability testing of FRPs should there be any limitations. *J Compos Construct* 2010;14(4):361–7.
- [26] Benmokrane B, Robert M, Mohamed H, Ali AH, Cousin P. Durability assessment of glass FRP solid and hollow bars (rock bolts) for application in ground control of jurong rock caverns in Singapore. *J Compos Construct* 2016;06016002 [https://doi.org/10.1061/\(ASCE\)CC.1943-5614.0000775](https://doi.org/10.1061/(ASCE)CC.1943-5614.0000775).
- [27] Benmokrane B, Ali AH, Mohamed HM. Durability of glass FRP solid and hollow bars (rock bolts) for the application in ground control of the jurong rock caverns in Singapore. QC, Canada: Univ. of Sherbrooke; 2012.
- [28] Benmokrane B, Ali AH, Mohamed HM, Safty A. Long-term tensile properties of carbon FRP cable. 15th international european bridge conference, London, UK 8–10th, July 2014. 2014.
- [29] Elgabbas F, Ahmed E, Benmokrane B. Physical and mechanical characteristics of new basalt-FRP bars for reinforcing concrete structures. *Construct Build Mater* 2015;95:623–35.
- [30] Sim J, Park C, Moon DY. Characteristics of basalt fiber as a strengthening material for concrete structures. *Compos Part B* 2005;36(6–7):504–12.
- [31] Parnas R, Shaw M, Liu Q. Basalt fiber reinforced polymer composites. Technical Rep. No. NETCR63, new england transportation consortium C/O advanced technology & manufacturing center. River, MA: Univ. of Massachusetts Dartmouth, Fall; 2007.
- [32] Pultrall. Composite reinforcing rods technical data sheet. Canada: Thetford Mines; 2015.
- [33] ASTM. Standard test method for transverse shear strength of fiber-reinforced polymer matrix composite bars. 2011. ASTM D7617, West Conshohocken, PA.
- [34] ASTM. Standard test method for flexural properties of fiber reinforced pultruded plastic rods. 2009. ASTM D4476, West Conshohocken, PA.
- [35] ASTM. Standard test method for apparent horizontal shear strength of pultruded reinforced plastic rods by the short beam method. 2008. ASTM D4475, West Conshohocken, PA.
- [36] ASTM. Standard test method for transition temperatures and enthalpies of fusion and crystallization of polymers by differential scanning calorimetry. 2012. ASTM D3418, West Conshohocken, PA.
- [37] Bulletin 40 Fib. FRP reinforcement in concrete structures. Lausanne, Switzerland: International Federation for 440 Structural Concrete; 2007.

- [38] ACI (American Concrete Institute). Guide test methods for fiberreinforced polymers (FRPs) for reinforcing or strengthening concrete structures. 2004. ACI 440.3R-04, Farmington Hills, MI.
- [39] ACI (American Concrete Institute). "Prestressing concrete structures with FRP tendons. 2004. ACI 440.4R-04, Farmington Hills, MI.
- [40] ACI (American Concrete Institute). Specification for carbon and glass fiber-reinforced polymer bar materials for concrete reinforcement. 2008. ACI 440.6M-08, Farmington Hills, MI.
- [41] ACI (American Concrete Institute). Guide for the design and construction of structural concrete reinforced with FRP bars. 2015. ACI 440.1R-15, Farmington Hills, MI.
- [42] Ali NM, Wang X, Wu Z, Hassanein AY. Basalt fiber reinforced polymer grids as an external reinforcement for reinforced concrete structures. *J Reinforc Plast Compos* 2015;34(19):1615–27.
- [43] ASTM. Standard test method for curing properties of pultrusion resins by thermal analysis. 2003. ASTM D5028, West Conshohocken, PA.
- [44] ASTM. Water absorption of plastics. 2010. ASTM D570, West Conshohocken, PA.
- [45] ASTM. Tensile properties of fiber reinforced polymer matrix composite bars. 2011. ASTM D7205, West Conshohocken, PA.
- [46] Benmokrane B, Ali AH, Mohamed HM, ElSafty A, Manalo A. Laboratory assessment and durability performance of vinyl-ester, polyester, and epoxy glass-FRP bars for concrete structures. *J. Compos.: Part B* 2017;114:163–74.
- [47] CSA (Canadian Standards Association). Specification for fibre reinforced polymers. 2010. CAN/CSA S807-10, Toronto.
- [48] Katsuki F, Uomoto T. Prediction of deterioration of FRP rods due to alkali attack. In: Taerwe L, editor. Proceedings of the second international RILEM symposium (FRPRCS-2), non-metallic (FRP) reinforcement for concrete structures. London: E&FN Spon; 1995. p. 83–9.
- [49] Shen CH, Springer GS. Moisture absorption and desorption of composite materials. *J Compos Mater* 1976;10:2–20.
- [50] Tannous FE, Saadatmanesh H. Environmental effects on the mechanical properties of e-glass FRP rebars. *ACI Mater J* 1998;95(2):87–100.
- [51] ASTM. Standard test method for compositional analysis by thermogravimetry. 2003. ASTM E1131, West Conshohocken, PA.
- [52] CSA (Canadian Standards Association). Design and construction of building structures with fiber reinforced polymers. 2012. CAN/CSA S806-12, Toronto.